

2025 CpSc H Q1

Section: Computer Systems

Topic: Data Representation (Two's Complement Range)

Question Summary

Given that the lowest representable value in two's complement is -512 for a fixed number of bits, (a) state the largest representable value; (b) state the number of bits used.

Worked Solution

Key fact: For an n -bit two's complement system, the range is from -2^{n-1} to $2^{n-1}-1$.

We are told the lowest number is -512 . Match this to -2^{n-1} :

$$-2^{n-1} = -512 \Rightarrow 2^{n-1} = 512 = 2^9 \Rightarrow n - 1 = 9 \Rightarrow n = 10.$$

So the number of bits is 10 bits.

The largest representable value is $2^{n-1} - 1 = 2^9 - 1 = 512 - 1 = 511$.

Also on this page: Digital signatures (Q2)

Two benefits you can cite:

- Authenticity — verifies the sender's identity (the signature was created with the sender's private key).
- Integrity — detects tampering (any change to the message invalidates the signature).
- Non-repudiation — the sender cannot plausibly deny sending the signed document.

Final Answer

(a) 511 (b) 10 bits. Two valid benefits of digital signatures include authenticity, integrity and non-repudiation (any two for full credit).

Revision Tips

- Memorise the two's complement range: -2^{n-1} to $2^{n-1}-1$.

- If given the minimum value, set -2^{n-1} equal to it and solve for n ; then compute the maximum as $2^{n-1}-1$.
- Digital signatures hinge on public-key cryptography: sender signs with their private key; recipients verify with the public key.
- For exam wording like “state two benefits,” write concise, distinct points (e.g., authenticity and integrity).

